



Grassland destruction causes shifts in plant traits that persist during recovery

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Old-growth grasslands assemble over millennia and support high biodiversity and many ecosystem services, but continue to undergo widespread destruction for production of crops, timber, and minerals. How biodiversity recovers after grassland destruction and the functional consequences of this recovery can be informed by plant functional traits—i.e., morpho-physiological features that influence growth, survival, and reproduction. Yet we lack understanding of how functional traits differ between old-growth and secondary (i.e., recovering) grassland plant species at the global scale, hindering understanding of community assembly processes that transcend geographical regions. Addressing this gap, we synthesized trait data for 742 plant species from 24 studies across six continents to understand how plant species indicative of old-growth and secondary grasslands compare in their traits. Compared to old-growth grasslands, plants indicative of secondary grasslands were taller and had leaves with more nitrogen content, higher specific leaf area, and lower dry-matter content. Notably, these trait differences that make old-growth grassland plants more resource conservative and secondary grassland plants more resource acquisitive persisted over a century. Taken together, these findings help to explain why many secondary grasslands support altered plant communities for decades to centuries and emphasize the importance of old-growth grassland conservation in the light of the unique plant forms and functions they support. Persistent trait shifts associated with grassland destruction could accelerate nutrient cycling, and our findings provide guidance for recovering old-growth grassland biodiversity through ecological restoration.

old-growth grassland | savanna | biodiversity | functional traits | community assembly

Human activities are altering most ecosystems at an unprecedented scale, and these changes are particularly evident in grasslands (1). Grasslands are undergoing two kinds of land-use changes. First, old-growth grasslands that have assembled over millennia continue to be destroyed for production of crops, timber, and minerals (2–4). Second, cessation of intensive land uses like row-crop cultivation across millions of hectares worldwide provide opportunities for biodiversity recovery and the development of secondary grasslands (5, 6). These patterns of conversion and abandonment are among the most widespread forms of land-use changes on Earth, and yet how old-growth and secondary grasslands differ ecologically remains poorly understood—endangering the high biodiversity and ecosystem functioning of one of the largest terrestrial biomes (3, 7).

Old-growth and secondary grasslands often support very different biodiversity levels, but it remains unclear why these differences exist (2, 6). Secondary grasslands can not only contain fewer but also distinct plant species than the old-growth grasslands they replace (reviewed in ref. 8). For example, many native perennial grasses and forbs that are characteristic of old-growth grasslands can be missing or replaced by non-native weedy annuals in secondary grasslands, particularly in the initial decades of recovery (e.g., refs. 9 and 10). Once destroyed, plant diversity of many old-growth grasslands is usually slow to reassemble (11–13). But we do not fully know why we see these patterns, and what morphological and physiological features or traits enable certain plant species to reestablish in secondary grasslands and not others. This gap in our understanding continues to impede us from formulating general “rules” in community ecology that are independent of species’ taxonomic identities (14). Consideration of plant functional traits—a common currency comparable across species and the wide bioclimatic gradients occupied by grasslands—might offer a mechanistic explanation for why we see such different communities (14).

Significance

Destruction of old-growth grasslands due to human land uses is contributing to the global biodiversity crisis. Yet, our understanding of why some plant species recover after destruction and others do not is limited. Analyzing data for 742 plant species, we show that functional traits—morpho-physiological features that influence growth, survival, and reproduction—serve as a missing link to understand why some species recover and not others. Compared to old-growth grasslands, plants indicative of secondary (i.e., recovering) grasslands had traits that enabled rapid acquisition of light, water, and nutrients, and these trait differences persist for over a century. Such trait shifts likely impact ecosystem functioning, inform restoration efforts, and underscore the need to conserve old-growth grasslands for their functionally unique plant forms.

The authors declare no competing interest.

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This is because plant functional traits influence the establishment, growth, survival, and reproduction of a species in a given environment and therefore are thought to mediate community reassembly including in secondary grasslands (15).

We expect plants indicative of old-growth and secondary grasslands to differ in several ways with respect to their traits. First, many plants indicative of old-growth grasslands would be native perennials, while those indicative of secondary grasslands would be non-native annuals (2, 16). Second, old-growth and secondary grassland plants may differ across two major functional-trait axes (17). One axis of global plant trait variation corresponds to a trade-off between leaf construction cost and carbon assimilation (i.e., the leaf economic spectrum; ref. 18). On one end of this spectrum are species with leaves with: low specific leaf area (SLA), low nitrogen concentrations, high leaf dry matter content (LDMC), long lifespan, and low photosynthesis rates (18). These “slow” species use resources conservatively and are best adapted to nutrient-poor environments (19). On the other end of this spectrum are species with leaves with: high SLA, high nitrogen concentrations, low LDMC, short lifespan, and high photosynthesis rates (18). These “fast” species are acquisitive in their resource use and are best adapted to fertile environments (19). Across the leaf economic spectrum axis, we expect plants indicative of old-growth

grasslands to be conservative, while plants indicative of secondary grasslands to be acquisitive because secondary grasslands are often nutrient-enriched as a legacy of past cultivation (e.g., ref. 20) and expected to contain more non-native and generalist species, which are often “fast” (2, 21–24; Fig. 1 and *SI Appendix*, Table S1). A second axis of trait variation corresponds to the correlation between plant size and plant organs. Taller plants often also have larger leaves and seeds and vice-versa for shorter-statured species (25, 26). Taller, large-leaved, and large-seeded plants are competitive and dominate in nutrient-rich environments (27), such as those present in secondary grasslands. Across the size axis, compared to plants indicative of old-growth grasslands, we expect those indicative of secondary grasslands, such as non-native and generalist species, to be competitive: i.e., taller, with larger leaves and larger seeds; although there is considerable variation in seed mass at any given height, therefore, the differences in seed mass between the two grassland types could likely be modest (12, 17, 22, 25, 28; Fig. 1 and *SI Appendix*, Table S1). The size axis for above-ground traits is also related to belowground traits. For example, plant height is typically negatively related to root-to-shoot biomass ratio in grassland plants (29). Indeed, many plants indicative of old-growth grasslands have considerable biomass belowground in the form of storage organs—that provide an advantage in repeatedly grazed

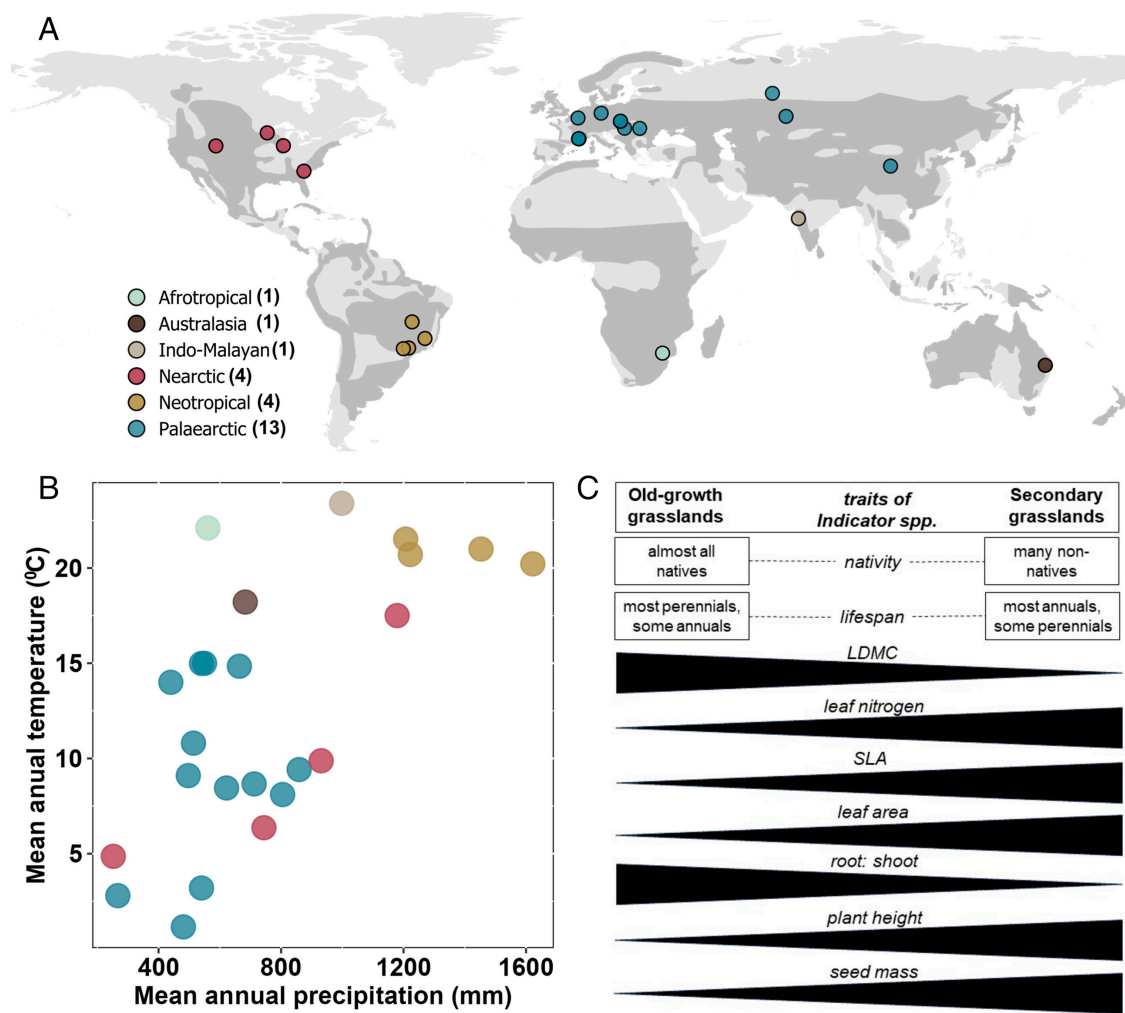


Fig. 1. Locations of the $n = 24$ studies included, the climatic envelope they occupy, and hypotheses that we test. (A) Distribution of study sites across biogeographic realms. Numbers in parentheses after the realm names show number of studies included in that realm. Approximate extent of global grasslands (including savannas and open-canopy woodlands), depicted in dark gray (data from ref. 7). (B) Bioclimatic envelope occupied by the sites. (C) Hypothesized trait differences between plants indicative of old-growth and secondary grasslands (key references used: 2, 12, 16–18, 22, 23, 25, 27, 28, and 30; also see *SI Appendix*, Table S1).

and burnt habitats, suggesting a high root-to-shoot ratio (2, 16, 30). Yet, across grasslands globally, which differ in several ways, we do not know whether plants indicative of old-growth and secondary grasslands occupy different trait spaces.

Several factors underscore the need for a global synthesis of how plants indicative of old-growth and secondary grasslands differ in their traits. Earth's grasslands occur across a broad range of climates and disturbance regimes, support regional floras with distinct ecological and evolutionary histories, and have undergone different kinds of destruction (e.g., mining, tillage agriculture, and afforestation)—all of which could modify responses of traits to land-use change (3, 4, 31). While emerging research suggests some of our expectations about trait differences might be true for some grasslands (e.g., refs. 21 and 23), it is imperative that we test if these expectations hold true across different global grasslands irrespective of the taxonomic identities of the species. A global synthesis will thus allow us to test for patterns that transcend species' taxonomic identities. Additionally, secondary grasslands worldwide span a large range of recovery ages, from recently abandoned fields after intensive cultivation to those abandoned decades or centuries ago, which could also contribute to trait differences between the two grassland types. For example, if secondary grasslands are recovering toward old-growth conditions, we might expect to see plant traits converging over time (e.g., ref. 32).

In this study, we quantify for the first time the functional trait space of plants indicative of old-growth and secondary grasslands worldwide (Fig. 1 and *SI Appendix*, Table S1). We asked: Q1) How do plants indicative of old-growth and secondary grasslands differ in their lifespan and nativity? Q2) How do plants indicative of old-growth and secondary grasslands differ in their functional traits? Specifically, we quantified how plants indicative of the two grassland types differ a) across key axes of plant-trait variation (i.e., suites of traits), b) with respect to nativity and plant functional types and how these relate to the suites of traits, and c) in individual traits? Q3) Do traits of plants indicative of old-growth and secondary grasslands converge over time, during recovery?

To address these questions, we first collated ground-layer plant community composition data for studies at 24 sites—each with paired old-growth and secondary grasslands—across six continents. These sites spanned a wide bioclimatic envelope ranging from 252 to 1,622 mm mean annual precipitation (MAP) and 1.2 to 23.4 °C mean annual temperature (MAT; Fig. 1 *A* and *B* and *SI Appendix*, Table S2). From these data, we identified a set of indicator species for each of the two grassland types for each site (742 species total; *SI Appendix*, Table S3). An indicator-species approach was ideal for testing hypotheses about the traits of species indicative of old growth vs. secondary grasslands (Fig. 1C; 2). We believe that our focus on indicator species is also timely given how widely conservation scientists and restoration practitioners continue to rely on them (e.g., refs. 2, 6, 9, 33, and 34). A strength of the indicator-species approach is that it acknowledges the contribution of not just the most abundant species that characterize a particular land use, but also the contribution of rare or infrequent species that often get ignored in abundance-based approaches. For example, our indicator species analyses identify species that are both locally dominant and therefore contribute most to community-level functioning (e.g., several perennial grasses from India, Australian, and Brazilian savannas, including the iconic “big four” prairie grasses of the United States), and also species that are locally or globally rare and uncommon yet of high conservation and restoration value. For all the indicator species, to inform two main axes of plant trait variation, we collated data for seven continuous traits [LDMC, SLA, leaf area (LA), leaf nitrogen (leafN), plant height, seed mass, root-to-shoot ratio; 4,944 measurements] alongside plant nativity, lifespan, and functional

types. Finally, we investigated if trait space occupied by indicator plants of the two grassland types changed over recovery age (ranging from 1 to 300 y) using a subset of studies. These findings provide a mechanistic understanding of the consequences of widespread grassland destruction by predicting trait shifts in global grasslands, with implications for ongoing efforts to restore lost biodiversity.

Results

Q1: Grassland Destruction Results in a Conspicuous Loss of Perennials and a Relatively Modest Loss of Native Indicator Species. Compared to old-growth grasslands, secondary grasslands supported a lower proportion of indicator plant species that were perennial (mean perennial indicator species for old-growth grasslands and secondary grasslands = 82.9% and 59.7%, respectively; Wilcoxon test, $P = 0.0061$; *SI Appendix*, Fig. S1A). We further found that among the perennial indicator species that showed this decline, perennial graminoids and subshrubs contributed the most, while perennial forbs were not significantly affected by grassland destruction (*SI Appendix*, Fig. S1D, F, and G). Concerning nativity, old-growth grassland destruction led to a modest drop in the proportion of indicative species that were native (mean native species for old-growth grasslands and secondary grasslands = 99.3% and 87.9%, respectively; Wilcoxon test, $P = 0.0059$; *SI Appendix*, Fig. S1B).

Q2: Plants Indicative of Secondary Grasslands Have Traits That Make Them More Resource Acquisitive and Competitive than Those of Old-Growth Grasslands. Multivariate analysis of the six core traits revealed that plant species indicative of old-growth and secondary grasslands occupied distinct trait spaces along the leaf economic spectrum (i.e., PC1 of the global PCA): Species indicative of secondary grasslands had leaves with more nitrogen content, higher SLA and lower LDMC compared to species indicative of old-growth grasslands (Cliff's delta = -0.332 , Wilcoxon test, $P < 0.0001$; PERMANOVA, $R^2 = 0.031$, $P = 0.001$; Fig. 2A). In line with our predictions, species indicative of old-growth grasslands were more resource-conservative (i.e., “slow”), while those indicative of secondary grasslands were more resource acquisitive (i.e., “fast”; Fig. 1C). Sensitivity analyses confirmed that these results of differences in trait suites were qualitatively similar across conservative and liberal sensitivity tests across different levels of data inclusion (*SI Appendix*, Figs. S3–S6). Plant species indicative of the two grassland types also differed in their patterns of dispersion within trait space, hence we interpret the PERMANOVA results with caution and as but one line of evidence alongside others (e.g., Cliff's delta and the univariate analyses). Compared to secondary grasslands, old-growth grassland plant species—while more resource conservative on average—occupied a larger trait space that also included many resource acquisitive species (average distance to group's spatial median for old-growth and secondary grasslands = 2.339 and 2.168, respectively; $P = 0.044$; Fig. 2A).

Plant nativity (PERMANOVA, $R^2 = 0.010$, $P = 0.008$) and functional types (PERMANOVA, $R^2 = 0.11$, $P = 0.001$) explained additional variation in the multivariate trait space. Non-native plant species were more resource acquisitive compared to natives (Fig. 2C). Annual forbs were resource acquisitive, while perennial graminoids and subshrubs were resource conservative (Fig. 2B and C).

The univariate pairwise analyses provided further insights and corroborated the results from the multivariate analyses. We found similar patterns of trait differences for the three leaf economic spectrum traits (i.e., SLA, LDMC, leafN) as we found in the

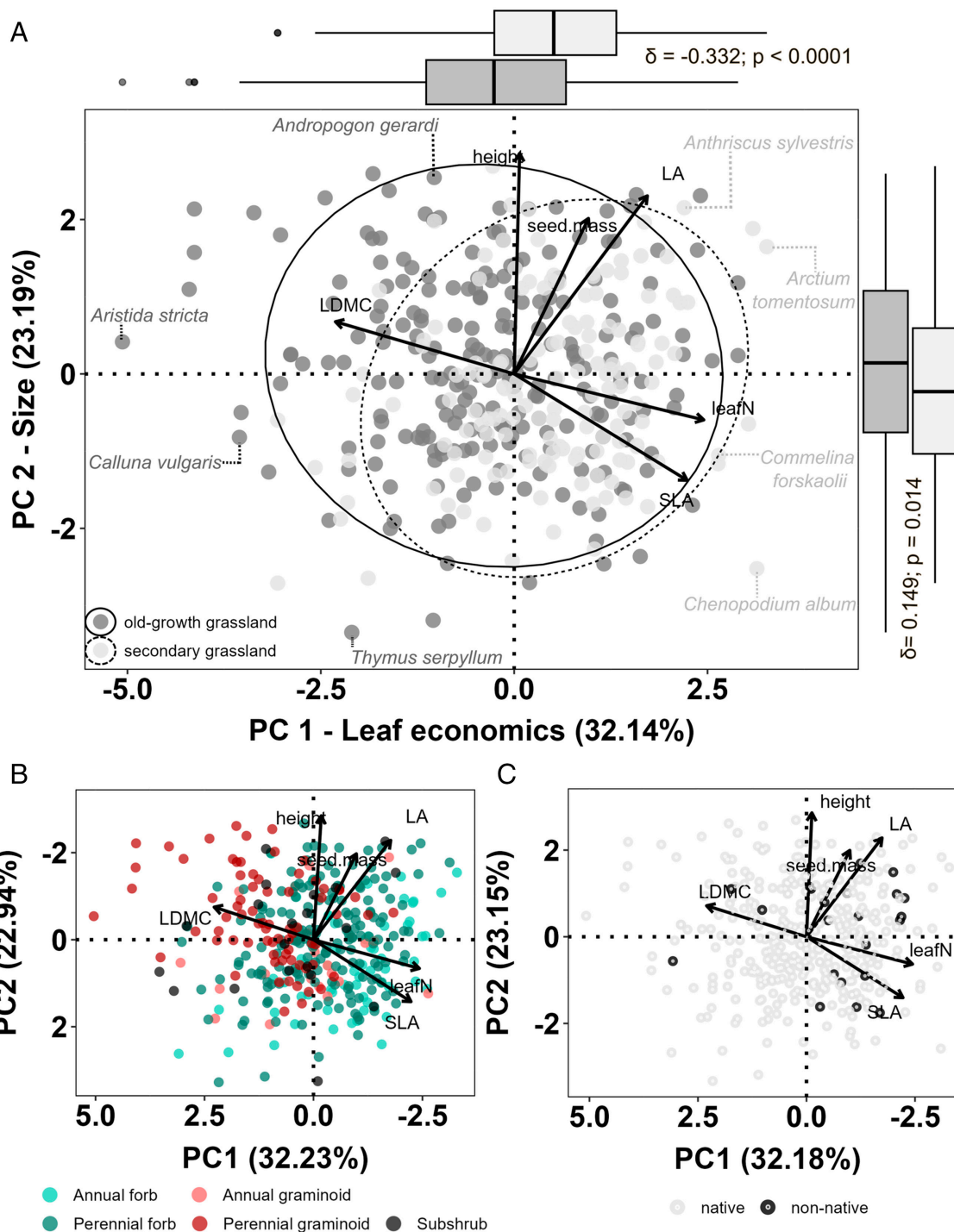


Fig. 2. Grassland destruction results in a shift in trait suites. (A) Multivariate trait space occupied by indicator species of old-growth (226 species) and secondary grasslands (150 species) for which data for all six core traits (solid arrows; LDMC, SLA, LA, leafN, plant height, and seed mass) were available. Solid and dashed circles represent 95% confidence ellipses for two grassland types. Centroids of trait space occupied by indicator plants of two grassland types differ (PERMANOVA, $P = 0.001$). Box plots near the top and right margins are PC1 and PC2 scores respectively (color coded by grassland type). Cliff's delta (δ) quantifies the strength of differences, and P -values are for the Wilcoxon rank-sum tests. Four representative species for each grassland type occupying the edges of the trait space are labeled. (B) Indicator species, when grouped by plant functional types occupy distinct multivariate trait space (C): non-native and native indicator species differ in their trait suites.

multivariate analyses (Fig. 3 A–C). However, differing from the multivariate results, for the size and competition-related traits, the univariate analyses showed that secondary grassland plant species were taller (and therefore more competitive) than those of old-growth grasslands (Fig. 3F). Seed mass, LA, and root:shoot did not differ between grassland types (Fig. 3 D, E, and G). In supplemental analyses we found that the difference in SLA between plant indicators of secondary and old-growth grasslands

was larger in wet tropical grasslands than in drier temperate ones ($R^2 = 0.27$, $P = 0.018$; SI Appendix, Fig. S7).

Q3: Trait Differences Persist for Over a Century. The trait space occupied by indicator plant species showed subtle yet consistent differences between the two grassland types in each of the three age-classes [young (1 to 9 y), PERMANOVA, $R^2 = 0.055$, $P = 0.013$; intermediate (10 to 90 y), PERMANOVA, $R^2 = 0.069$, $P = 0.003$;

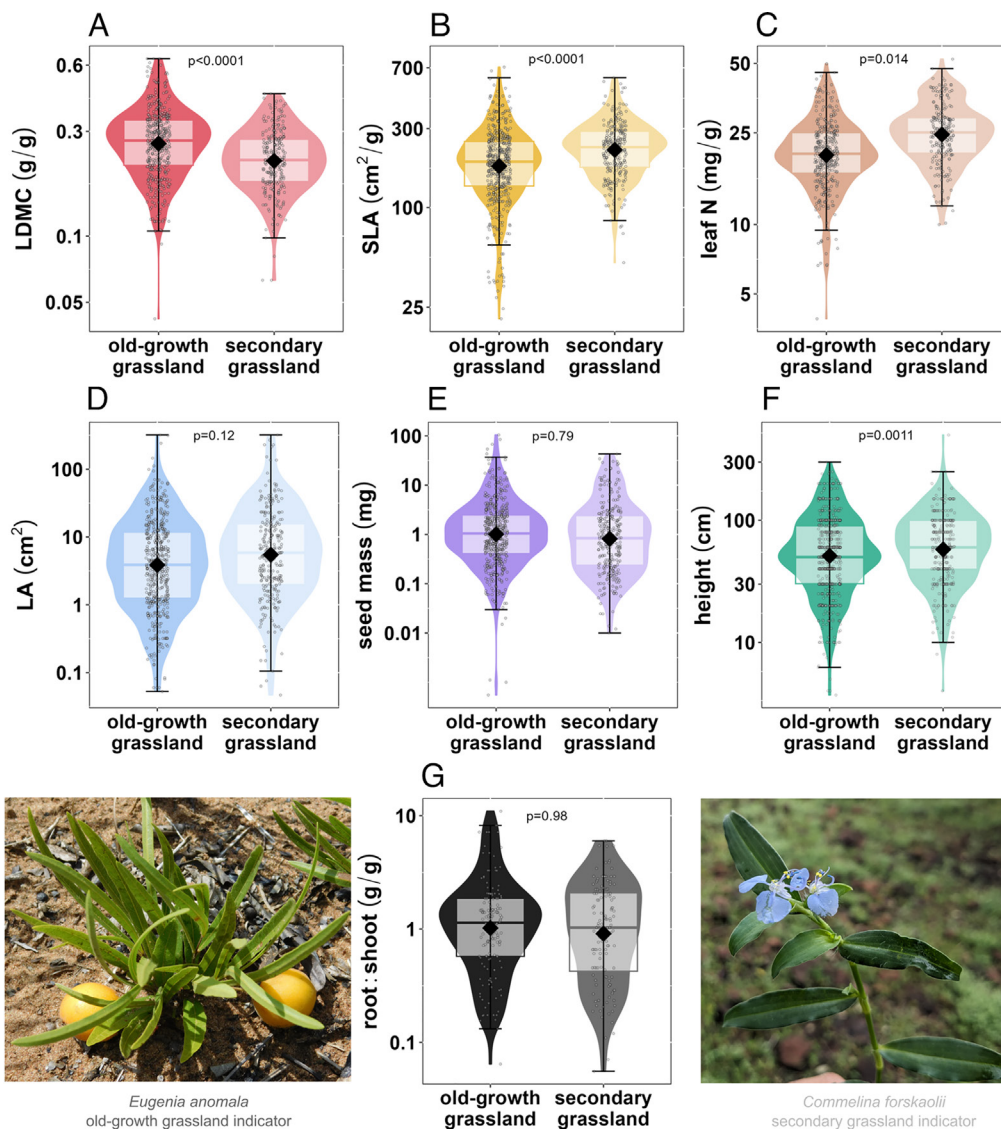


Fig. 3. Compared to old-growth grasslands, plants indicative of secondary grasslands have traits that make them more resource acquisitive and competitive. (A–G) Univariate pairwise trait differences between old-growth and secondary grasslands for seven continuous traits (LDMC, SLA, LA, leafN, plant height, seed mass, and root:shoot). For (A–G), the horizontal line represents the median; whiskers extend up to 1.5 times the interquartile range; diamonds indicate means; *P*-values are for the mixed models that account for site and species as random effects. Trait values are log₁₀ transformed.

old (300 y), PERMANOVA, $R^2 = 0.072$, $P = 0.019$; Fig. 4A and B]. For the young age-class, indicators of secondary grasslands differed from indicators of old-growth grasslands across the size axis (i.e., PC2; Fig. 4A): with secondary grassland plant species being taller, with larger leaves, and heavier seeds (Fig. 4B). For the intermediate age-class, indicators of secondary grasslands differed from indicators of old-growth grasslands primarily across the leaf-economic spectrum axis, but also to some degree on the size axis (Fig. 4A): with secondary grassland plant species having leaves with more nitrogen content, higher SLA, lower LDMC, and greater LA (Fig. 4B). For the old age-class, indicators of secondary grasslands differed from indicators of old-growth grasslands across the size axis (i.e., PC2; Fig. 4A): with secondary grassland plant species being markedly taller, but similar in all other traits (Fig. 4B).

Discussion

Our global dataset reveals that grassland destruction results in plant species with traits that make them more resource-acquisitive and competitive compared to plant species indicative of old-growth

grasslands (Figs. 2A and 3). Because the functional traits we studied influence community assembly (15, 35), these findings serve as one of the missing pieces in our understanding of why old-growth and secondary grasslands continue to show distinct plant diversity and composition over time. This missing piece has taken us a step closer to formulating general mechanisms of grassland community assembly that transcend specific sites and species pools enabling us to draw conclusions on much larger spatial (six continents) and temporal scales (up to 300 y). Further, the long recovery time of old-growth grassland functional traits helps explain why many secondary grasslands support altered biodiversity for over a century (e.g., ref. 13; Fig. 4). In light of this slow recovery and functional ramifications of destruction, one thing is clear—halting the destruction of old-growth grasslands is imperative (2, 3, 9).

Our findings supported many, but not all, of the hypothesized trait differences between old-growth and secondary grasslands (Fig. 1C). In line with our predictions, secondary grasslands supported fewer indicative species that were native and perennial, and many indicative species with more resource acquisitive and competitive traits: i.e., greater SLA, leafN, height, and lower LDMC

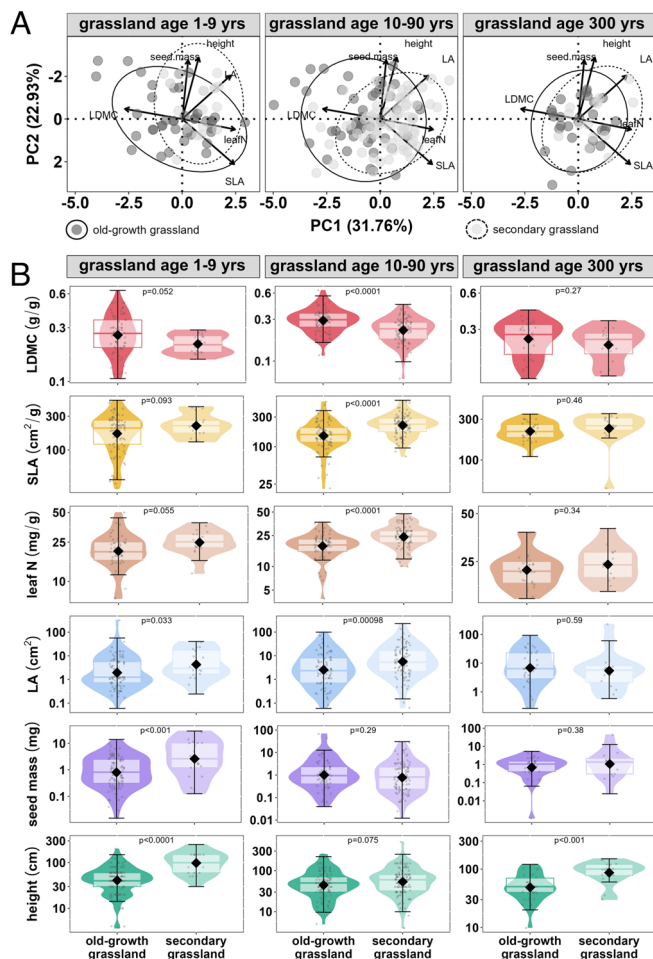


Fig. 4. Many differences in trait suites and traits persist during recovery. (A) Multivariate trait differences between the indicator species of old-growth and secondary grasslands over three age-classes: 1 to 9 y (4 studies), 10 to 90 y (6 studies), and 300 y (1 study) for six core traits (solid arrows; LDMC, SLA, LA, leafN, plant height and seed mass). Indicator species of the two grassland types differ in their multivariate trait space for all age-classes (PERMANOVA; $P < 0.05$). (B) Univariate trait differences across three age-classes for the six core traits. The horizontal line represents the median; whiskers extend up to 1.5 times the interquartile range; diamonds indicate means. Trait values are $\log_{10}$ transformed.

(Figs. 2A and 3A–C and F and SI Appendix, Fig. S1A and B). These trait differences highlight the predictable role of leaf-economic spectrum traits in mediating community assembly of grasslands—much like assembly of forests—making a case for the wider incorporation of leaf economic traits in studies of grassland ecology and restoration, where large proportions of variations in community assembly often remain unexplained (36, 37). Contrary to our predictions, we did not detect differences for LA and seed mass, suggesting successful reestablishment of species with values similar to old-growth grassland species for these traits during recovery (Figs. 3D and E and 4B). High variability in LA and seed mass globally, and in our dataset, where they span over four orders of magnitude, could also underpin some of these patterns (17; Fig. 3D and E). We also did not detect differences in root:shoot, contradicting our expectations (Fig. 3G). This could be because of the large proportion of annuals in the dry Mediterranean old-growth grasslands (SI Appendix, Fig. S1A), or incomplete destruction of belowground storage organs during land use prior to abandonment, or an artifact of the relatively fewer measurements for this trait. Our findings are broadly similar to those from forests, where recovering habitats also support more species with acquisitive traits, compared to remnants (37).

Our synthesis spans biogeographically and ecologically diverse grasslands across six continents (Fig. 1A and B) to provide new insights. Among these, our study uncovers a remarkable diversity of plant phenotypes indicative of global old-growth grasslands, reflecting the myriad ways in which plants adapt to diverse environments (Fig. 2A). While on average, plants indicative of old-growth grasslands are resource-conservative, shorter, and perennial, other plant forms were also well represented (Fig. 2A and SI Appendix, Fig. S1A). For example, old-growth plant species include tall (and therefore competitive) C_4 perennial grasses of fire-prone, megafauna-grazed ecosystems such as North American prairies (e.g., *Andropogon gerardi* and *Panicum virgatum*), South African veld, Australian savannas (e.g., *Themeda triandra*) and Indian savannas (e.g., *Cymbopogon martinii*). Old-growth plant species also include those with resource acquisitive traits, such as annual forbs typical of the dry and grazed Mediterranean steppe (e.g., *Euphorbia falcata* and *Erodium cicutarium*) and perennial forbs of Czech's hyperdiversity-rich White Carpathian Mountain grasslands (e.g., *Salvia pratensis*). While most old-growth indicators are nonwoody, woody shrubs are important components of two global biodiversity hotspots—fire-dependent longleaf pine savannas of the southeast United States (e.g., *Gaylussacia dumosa*) and the Brazilian Cerrado (e.g., *Eugenia anomala*; Fig. 2A and B). Our study, thus points to the wide functional trait space occupied by the plant species indicative of old-growth grasslands (Fig. 2A). Similarly, our results provide a compelling reason to reconceptualize the species pool of secondary grassland indicator species. By showing that indicative species of secondary grasslands consist of a mix of native and non-native species we clarify that the old-growth vs. secondary distinction is not synonymous with native vs. non-native, respectively (SI Appendix, Fig. S1B). This is because several natives in our dataset are functionally similar to non-natives in secondary grasslands (Fig. 2C; e.g., ref. 38).

Functional Traits Explain Biodiversity Loss Following Grassland Destruction. Our trait-based analyses provide insights into why some plant species indicative of old-growth grasslands may not reestablish in secondary grasslands, why other plant species that are not part of the old-growth grassland species pool establish in secondary grasslands, and how these changes eventually lead to an incomplete recovery of grassland biodiversity.

Three barriers may lead to an incomplete recovery of plants indicative of old-growth grasslands. First, seeds of target old-growth grassland species may not reach secondary grasslands because of dispersal limitation (39, 40). Focusing on two traits related to dispersal in our analyses, we found that indicator plant species reestablishing in secondary grasslands are similar in their seed mass but taller than those of old-growth grasslands (Fig. 3E and F). This suggests that postdestruction, the dispersal and recovery of plants indicative of old-growth grassland may not be constrained by seed mass alone, but rather by additional traits including height—agreeing with syntheses showing that plant height explains most while seed mass explains very little additional variation in dispersal distances (41). How other related aspects of dispersal relate to recruitment remain to be investigated, such as dispersal syndromes, seed longevity, and landscape context (11, 42, 43). Second, if seeds of old-growth grassland plants arrive in secondary grasslands, abiotic/environmental conditions altered during the destruction process, e.g., soil nutrients, pH, and microbiota, might prevent their establishment. Indeed, many secondary grasslands in our analyses had altered soil nutrients (e.g., refs. 10 and 44–46). Secondary grasslands might also have altered historical disturbance regimes such as suppression of fire and/or grazing, which could limit establishment of old-growth grassland plants

via light limitation (20, 47, 48). Finally, even if old-growth grassland plants establish in secondary grasslands, given their low competitive ability (i.e., shorter height; Fig. 3F) and slow growth, they may not persist because of competition from fast growing plants with acquisitive traits (e.g., high SLA, high leafN, low LDMC; Fig. 3) that were not originally part of the old-growth grassland indicator plant species pool, that reestablish more quickly in secondary grasslands. Our data imply that such competitive exclusion by species with acquisitive traits (e.g., SLA) might be strongest in the wet tropics, in part explaining the larger biodiversity losses reported in the tropics compared to the temperate regions after grassland destruction (SI Appendix, Fig. S7; refs. 8 and 48).

Implications of Old-Growth Grassland Destruction for Ecosystem Functioning, Ecological Restoration, and Biodiversity Conservation. Many indicator species in our analyses are also locally dominant, prompting us to hypothesize how trait shifts at the species-level that we document (Fig. 3) could translate to shifts in ecosystem functioning. In doing so, we acknowledge that traits for species may not always mirror traits of communities, but in supplementary analyses show that they do so reasonably well particularly for traits associated with leaf economics in 15 of our studies (SI Appendix, Table S5). Our findings that old-growth grassland destruction favors plants that are characterized by resource-acquisitive traits has implications for nutrient cycling in global grasslands (Figs. 2A and 3). Compared to old-growth grasslands, we expect secondary grassland communities to show accelerated rates of nutrient cycling (e.g., of nitrogen and phosphorus), which in turn may facilitate “fast” species, further promoting eutrophic conditions (analogous to the roles of non-native species; refs. 49 and 50). Positive feedbacks between plant communities, soil attributes, and disturbance regimes may together reinforce the persistence of secondary grassland communities as alternate stable states, with direct implication for ecological restoration (51).

In the light of the large-scale destruction of old-growth grasslands, our results provide important implications for ecological restoration. Passive restoration, i.e., natural recovery with minimal human interventions, might not be enough to restore the lost plant communities of old-growth grasslands—our data points at the incomplete recovery of many indicator plants of old-growth grasslands even after a century of passive recovery (Fig. 4A and B). Such slow rates of passive recovery emphasize the need for active restoration, through interventions like seed sowing, production, and transplantation of target species, reinstating endogenous disturbances and altering abiotic conditions (40, 52, 53). While some active restorations succeed, at least in part by reintroducing the diversity of target old-growth grasslands (e.g., ref. 54), mimicking functional trait space in restorations remains challenging. For example, restored grasslands are often overwhelmingly seeded by fast-growing resource-acquisitive plant species, making them functionally distinct from the target old-growth grasslands dominated by resource-conservative species, likely with cascading impacts on nutrient cycling, pollinators, carbon sequestration, and productivity (e.g., refs. 23 and 55). In this context, our findings suggest that active restoration approaches must consider the wide, but largely conservative functional trait space occupied by indicator species of old-growth grasslands (Figs. 2A and 3). This could be achieved by designing site-specific propagule mixes in a way that matches the trait space pool of plants indicative of one or more old-growth grassland sites that serves as a reference to a restoration site (56). Doing so would require greatly expanding the “restoration species pool”—the set of species available from producers, a key bottleneck for most restorations—by explicitly taking into account functional traits (33, 34).

Finally, our findings underscore the critical importance of halting the destruction of old-growth grasslands, in the light of the high biodiversity and unique plant forms and functions they house (2, 3, 57; Figs. 2A and 4A; also see ref. 58). Conserving this vast reservoir of plant phenotypes first requires better acknowledging the distinction between old-growth and secondary grasslands, which, as we show, differ in several ways (SI Appendix, Figs. S1–S3). This distinction matters because of the very different conservation values of the two grassland types, differences in functionally important plant traits, differences in ecosystem functioning and because many old-growth grasslands continue to be misinterpreted as secondary, such as those in the tropics (e.g., refs. 59 and 60). Clearly, this distinction is more than a semantic issue, with major implications for both advancing fundamental grassland ecology and conserving global biodiversity.

Materials and Methods

Study Inclusion Criteria and Database Creation. To identify studies that match our requirements, we conducted a Web of Science search (code in SI Appendix, Supplementary Text) and included peer-reviewed journal articles based on specific inclusion criteria (see ref. 8 for details). Briefly, these were: 1) the study was conducted in a grassland, broadly defined as ecosystems supporting a continuous or near-continuous herbaceous ground layer, including savannas and open-canopy woodland. This includes a large number of herbaceous-dominated vegetation types, including some that are termed “woodlands” (e.g., Eucalypt woodlands of Australia, Indian dry forests/woodlands, and long-leaf pine woodlands of the United States; 61); 2) included old-growth grasslands—i.e., remnants without any previous large-scale exogenous human disturbance but continued endogenous disturbances such as fire and/or grazing (2, 53). We define old-growth grasslands broadly to include modest departures from the historical endogenous disturbance regimes. As such, we consider grasslands to be old-growth where lightning-ignited fires have been replaced by prescribed fires, where native megafauna have been replaced/supplemented by domestic livestock in moderate densities, and where mowing and/or haying serves as a surrogate for frequent above-ground disturbance (2). Our broad definition of old-growth includes both grasslands where soils or climate limit trees as well as those grasslands where fire and megafaunal herbivory limit trees (7, 62); 3) included secondary or “new” grasslands—i.e., grasslands formed postdestruction of old growth grasslands due to exogenous disturbances like tillage agriculture, mining, tree plantations, or other such major structural or functional changes (2, 6). We did not include study sites that were fertilized after abandonment (e.g., with nitrogen or phosphorus). Because our aim was to understand “passive” restoration in secondary grasslands, we omitted plots under “active” restoration treatments, such as plant reintroductions by seeds; 4) included plant community data for the ground-layer (i.e., understory) plants for the two grassland types.

We divided the Web of Science search into two steps, and we direct the readers to SI Appendix, Supplementary Text for the search code, screening, and filtering steps. Through our screening and filtering process, we were finally able to include a total of 24 studies in our analyses (SI Appendix, Fig. S2 and Table S2).

Identifying Indicator Species. For the final set of 24 studies, we obtained a site by plant species matrix associated with their publication, with sites designated as either old-growth or secondary grasslands. To ensure a single site by plant matrix per site, when needed we averaged multiple within-site plots to represent a single row for a single site. These matrices had either percent cover values or frequencies. We included only herbaceous species (i.e., graminoids, forbs, vines, and ferns) and woody subshrubs or shrubs (that were typically <2 m tall). Any woody species that typically grow >2 m tall were omitted. Then, for each dataset we ran an indicator species analysis to extract significant indicators ($P < 0.05$), specifying old-growth and secondary grassland types as groups, using the `multipatt` function in the `indicspecies` package (999 permutations; 63) in R v 4.3.1 (64). The indicator species analysis quantifies the associations of species patterns and combination of grouping treatments/sites to test for the best matching associations (63). In the light of large gaps in trait data for several species in the entire community composition datasets, using indicator species rather than entire communities offers a practical and

ecologically meaningful approach. The indicator species analyses returned 962 species in total across grassland types and studies (742 unique species, and the rest repeated) with 643 species for the old-growth grasslands, 319 for the secondary grasslands, and 110 shared between either grassland types or studies (*SI Appendix, Tables S2 and S3*). For each of these indicator species, we used the accepted scientific name (following ref. 65) and extracted the following categorical and continuous traits.

Trait Data: Categorical Traits. Nativity: we classified each indicator species as either native or non-native to the site using regional floras and supplemented some uncertain cases with data from ref. 65. Non-natives included species deliberately or accidentally introduced a few hundred or thousand years ago (e.g., *Arrhenatherum elatius* for Czech Republic) or a few decades ago (e.g., *Parthenium hysterophorus* for India).

Lifespan: We classified each indicator species as either perennial or annual following regional floras. We grouped biennials and annuals, to distinguish these from long-lived perennials.

Functional type: We classified each species as either: perennial graminoid, annual graminoid, perennial forb, annual forb, subshrub, and fern. Graminoids included members of Poaceae, Cyperaceae, and Juncaceae. We defined forbs as herbaceous nongraminoids including herbaceous vines and herbaceous legumes. Subshrubs (including small shrubs) were perennial plants <2 m tall with at least partly lignified (i.e., woody) stems.

Trait Data: Continuous Traits. We collated trait data from the following sources for seven continuous traits (i.e., LDMC, LA, SLA, leafN, height, seed mass, root-to-shoot ratio). We first prioritized site-level published and primary trait measurements, provided by coauthors. We defined plant height as the height of the tallest plant part from the ground—without distinguishing between vegetative or reproductive parts—as is typical for such global studies (e.g., refs. 24, 26, and 66). We sourced most data for maximum plant height from regional floras and provide a sensitivity analysis that reveals high correlation between maximum and vegetative plant heights—a pattern corroborated by other studies too (e.g., ref. 67; *SI Appendix, Fig. S8 and Table S2*). After site-level data, we included data from regional trait databases, and finally filled the remaining records from the global trait database TRY (68). By following this sequential approach for data collection, a strength of our synthesis is the large (mean = 56%, range = 15 to 100%) proportion of site and region-specific trait data, which helps minimize the confounding effects of phenotypic plasticity (*SI Appendix, Table S2*).

For data obtained from the TRY database, we used the *rtry* package to remove duplicated observations and those with a SD >4 from at least of the species/genus/family mean values (69). When multiple values were retrieved for a given species and trait from TRY, we averaged these, and assumed that interspecies trait variability will be greater than intraspecific variation as is typical for such syntheses (e.g., refs. 17, 66, and 70).

For our final trait by species matrix for the seven continuous traits, we thus collated 4,944 measurements out of the 6,734 possible measurements (i.e., had 26.6% missing values); for our final trait by species matrix for six core traits (i.e., excluding root:shoot), we could collate 4,656 measurements out of the 5,538 possible (i.e., had 16% missing values). We did not attempt to impute missing values given the emerging concerns that such imputation methods could compromise data quality (71).

Data Analyses.

Q1: Differences across nativity and lifespan. To understand how indicator species of the two grassland types differed in their nativity and lifespan, we first determined the proportion of native and perennial indicators for a given site and grassland type (the remainder were non-native and annual species, respectively). We then performed a paired Wilcoxon rank-sum test (because our data did not follow normal distribution) using the *stat_compare_means* function in the *ggpubr* package (72) to determine if indicators of the two grassland types differed in nativity, lifespan and repeated this analysis for the five functional types (i.e., annual forbs, perennial forbs, annual graminoids, perennial graminoids, and subshrubs; *SI Appendix, Fig. S1*).

Q2: Differences across trait suites and individual traits. We performed four complementary analyses (a to d) to understand how plants indicative of old-growth and secondary grasslands differ across traits. In doing so, we used

multivariate approaches to understand how indicator species differ across the suite of traits (a to c) and univariate approaches to understand how individual traits vary between the two grassland types (d).

a) To determine how indicator species of the two grassland types differed across the suite of traits, we performed a Principal Component Analysis (PCA) using the *prcomp* function, and visualized the first two Principal Components using the package *ggord* in R (73). To ensure the largest dataset without missing data, we focused on the six core traits for this analysis (LDMC, SLA, LA, leafN, plant height, seed mass), where we were able to get 61.6% of data (i.e., 2,256 measurements that had data for all six traits out of the 3,662 possible). For the PCA, we averaged trait values of indicator species that were repeated across studies within a grassland type so as to work with only a single row per species per grassland type (resulting in a matrix of 376 rows × 6 traits; i.e., 2,256 measurements). Trait values were $\log_{10}$ transformed and zero-centered following standard approaches (17, 74). To understand if the centroids of the trait space differed between the two grassland types, we performed a Permutational Multivariate ANOVA (PERMANOVA) using the *adonis2* function in the *vegan* package (75) and interpreted the results in the light of the results of test “b” below.

To understand how stable the results of this global PCA were, we performed four sensitivity analyses that presented two conservative (1, 2) and two liberal (3, 4) approaches in terms of data-volume (*SI Appendix, Figs. S3–S6*): 1) by removing 31 indicator species that were common to two grassland types; 2) by considering root:shoot trait alongside the six core traits (and thereby using a smaller number of rows with complete trait data for all 7 traits); 3) by removing leafN, the trait with lowest data entries among the six core traits (thereby increasing the number of dataset rows); 4) by considering all the species that were repeated across studies and grassland type (i.e., each species by study by grassland type combinations) which used the full dataset.

To understand how nativity and plant functional types can help explain additional variation in the multivariate trait space, we performed two more PCAs following the same steps outlined above to test if and how these categorical traits align with one or more trait suits (*Fig. 2 B and C*). Similar to the global PCA (*Fig. 2A*), where we averaged the trait values of species that repeated within a grassland type, these two PCAs also had one row for each species per grassland type per nativity-status (*Fig. 2B*) and per functional type (*Fig. 2C*).

b) To understand if the dispersion of the trait space differed between the two grassland types, we tested if the average distance of the indicator species to the group's spatial median differs between grassland types using the *betadisper* function in the *vegan* package (75). In cases where we detected significant differences in trait dispersion between the grassland types, we interpreted the PERMANOVA results with caution.

c) To understand how the indicator plant species of two grassland types differ in the scores of the first two Principal Components (PCs), we first performed a Wilcoxon rank-sum test for PC1 (i.e., the leaf economics axis) and PC2 (i.e., the size axis) scores for each grassland type and then used the package *effsize* to report Cliff's delta (8) associated with each test to quantify the magnitude of difference (76; *Fig. 2A*).

d) To understand how individual traits vary between the two grassland types, we performed pairwise univariate analyses for each of the seven continuous traits using the *lme4* package and graphed the data using the *ggplot2* package (77, 78). The multivariate analyses allowed only the inclusion of a subset of species with complete data for all six core traits, but these pairwise analyses utilize the full dataset enabling a robust trait-level comparison (84.1% of data; 4,656 measurements out of the 5,538 possible). Each of the seven mixed effects models consisted of the $\log_{10}$ transformed trait values as response variables, grassland type as predictor variables and two random intercepts, one for study and another for species. As supplementary analyses to our core questions, we quantified the contribution of biogeographical variables (latitude, longitude, MAP, MAT, and altitude) which might explain additional variation in trait differences between the two grassland types across sites (*SI Appendix, Fig. S7, Table S4, and Supplementary Text*).

Q3: Trait differences across time. To understand how indicator species' trait differences between the two grassland types changed over time, we first extracted the age of the secondary grassland for each study (i.e., the time since

abandonment). Then, we included studies that fit three age-classes based on the age of secondary grassland: young [1 to 9 y–community response to disturbance is markedly rapid up to decade in grasslands (e.g., refs. 79 and 80)], intermediate [10 to 100 y–community responses show much slower change per unit time after a decade of rapid change (e.g., ref. 13)], and old [>100 y–most grasslands at least need a century to recover lost biodiversity (e.g., refs. 8 and 13)]. Moreover, for studies that provided data for a broad range of secondary grassland age, we did not divide the data from such studies in more than one age category because that would have led to too few plots per age category and/or risk pseudoreplication. These criteria led to the inclusion of 11 studies across three age classes: young (1 to 9 y; 4 studies) intermediate (10 to 90 y; 6 studies), and old (300 y; 1 study). Similar to the steps outlined for the previous questions, to determine how the multivariate trait space for indicator species varied with age, we then ran a PCA for these 11 studies by assigning them to the three age-classes and ran a PERMANOVA to test if the centroids of the multivariate trait space differ between the grassland types across the three classes (and for the ease of visualization, we plot the results as three facets, each corresponding to one age-class). Last, analogous to the analysis for Q3, we also quantified the univariate pairwise differences for the six core traits across each of the three age-classes.

Data, Materials, and Software Availability. Data underlying the analyses are deposited in the GitHub Repository (81). No novel code was used during data analyses.

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